

Statistical Reasoning: Descriptive Statistics Project

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Project: Descriptive Statistics of the Percent Change of the Consumer Price Index for the Years 1974-1984

The dataset is treated as a sample.

Objectives:

- (1.) Describe the Dataset
 - (2.) Present the Dataset Using a Histogram
 - (3.) Determine the Measures of Center in the dataset
 - (4.) Determine the Measures of Variation (Spread) in the dataset
 - (5.) Determine the Measures of Location in the dataset
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Dataset Description:

The **Consumer Price Index (CPI)** is a measure of the average change over time in the prices paid by urban consumers for a market basket of consumer goods and services (U.S. Bureau of Labor Statistics). It is a metric used by the United States government to measure the effects of inflation on the price of consumer goods. The data I use is percent change year over year, so the price of the consumer basket increased by 11.0% in the year 1979.

Dataset:

Annual percent changes in selected Consumer Price Indexes for All Foods, 1974-1984 (Raw Data)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Annual percent changes in selected Consumer Price Indexes, 1974 through 2023																
2	Consumer Price Index item	1974	1975	1976	1977	1978	1979	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989
3	All food	14.3	8.5	3.0	6.3	9.9	11.0	8.6	7.8	4.1	2.1	3.8	2.3	3.2	4.1	4.1	5.8
4	Food away from home	12.7	9.4	6.8	7.6	9.1	11.1	9.9	9.0	5.4	4.4	4.2	3.9	3.9	4.0	4.1	4.6
5	Food at home	14.9	8.2	2.1	5.9	10.5	10.8	8.1	7.2	3.5	1.0	3.7	1.5	2.9	4.3	4.2	6.5
6	Meats, poultry, and fish	2.2	8.5	0.9	-0.6	16.7	14.9	3.7	4.1	4.1	-0.7	1.7	-0.4	4.4	6.5	3.5	5.0
7	Meats	1.8	8.5	0.2	-2.3	18.6	17.0	2.9	3.6	4.9	-1.2	0.3	-0.9	3.1	7.5	2.4	4.0
8	Beef and veal	2.9	1.0	-3.2	-0.7	22.9	27.4	5.7	0.8	1.4	-1.5	1.2	-2.1	0.6	7.6	5.5	6.4
9	Pork	-0.5	22.4	1.3	-5.4	12.9	1.6	-3.3	9.3	12.8	-0.9	-1.3	0.3	8.2	8.2	-3.0	0.6
10	Other meats	3.1	5.9	5.9	-0.6	17.7	14.7	3.8	4.3	3.0	-0.4	0.4	0.7	2.6	6.3	2.6	2.8
11	Poultry	-5.1	10.5	-4.1	0.7	10.4	4.9	5.2	4.1	-1.7	1.3	10.6	-1.0	7.5	-1.4	7.2	9.9
12	Fish and seafood	15.3	8.5	11.7	10.8	9.4	9.7	9.2	8.3	3.6	1.1	3.2	4.9	9.2	10.6	5.8	4.5
13	Eggs	0.4	-1.8	9.2	-3.2	-5.4	9.5	-1.8	8.2	-2.7	4.7	11.7	-16.6	6.8	-5.9	2.3	26.6
14	Dairy products	18.6	3.1	8.1	2.7	6.8	11.6	9.8	7.2	1.4	1.2	1.3	1.9	0.1	2.5	2.4	6.6
15	Fats and oils	41.9	10.7	-12.5	10.1	9.6	7.9	6.7	10.6	-2.7	1.4	9.4	2.2	-2.2	1.5	4.6	7.2
16	Fruits and vegetables	16.6	3.1	2.6	0.2	11.1	8.0	7.2	12.1	6.4	0.3	8.6	2.6	0.0	8.0	7.6	7.7

Dataset Link:

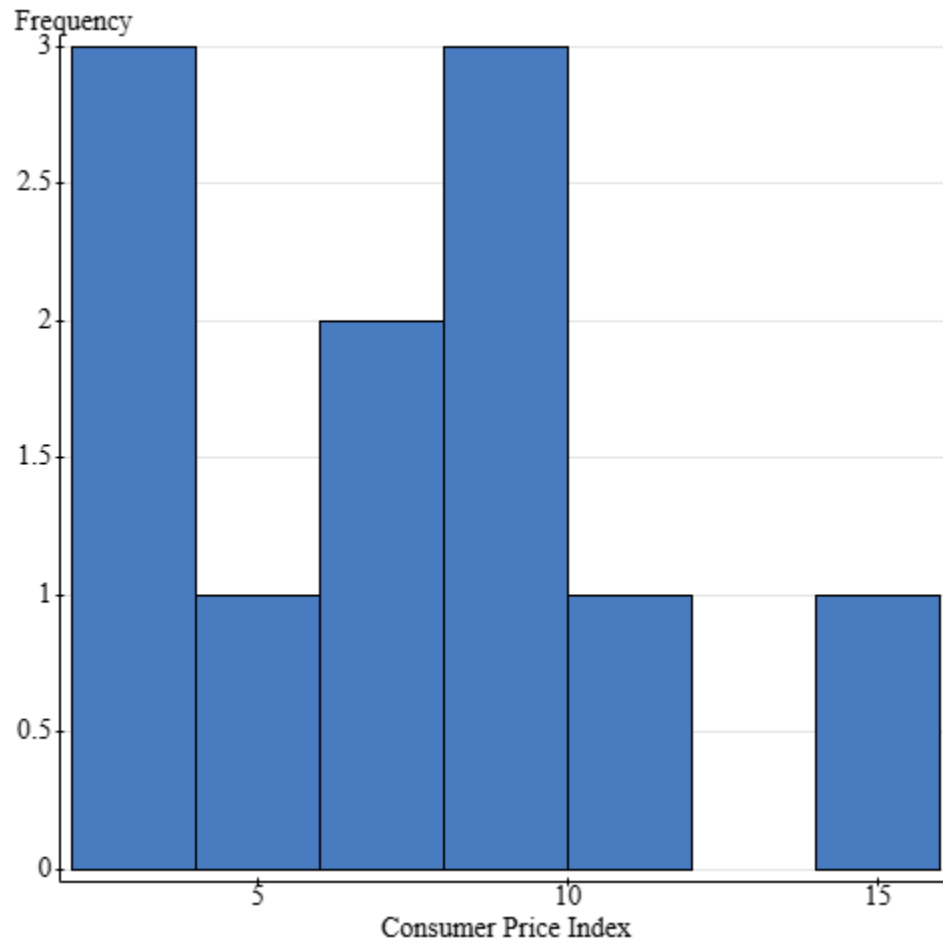
[Food Price Outlook - Catalog](#) (Once you access the link, scroll to “Downloads & Resources” and click the blue download button. Next, scroll down to “File Downloads” and click on “Consumer Price Index”. Next, click on “Download XLSX” for **Annual percent changes in selected Consumer Price Indexes, 1974 through 2023**. The specific data I used is from row “All Food” and column 1974-1984.

The dataset is

14.3%, 8.5%, 3.0%, 6.3%, 9.9%, 11.0%, 8.6%, 7.8%, 4.1%, 2.1%, 3.8%,

Technology:

All calculations were verified using a TI-84 Plus calculator provided by the BRCC Library and Pearson StatCrunch software.

Data Presentation:

The histogram has two mounds, clustered around [2,4) and [8,10). The three data points for [2,4) are 2.1%, 3.0%, and 3.8%. The three data points for [8,10) are 8.5%, 8.6%, and 9.9%.

Data Analysis:

I will be rounding to the hundredths place for all calculations.

Formulas:**Mean:**

$$\bar{x} = \frac{\Sigma Fx}{\Sigma F}$$

Midrange:

$$\text{Midrange} = \frac{\text{min} + \text{max}}{2}$$

Range:

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

Variance (Sample):

$$s^2 = \frac{\Sigma (x - \bar{x})^2}{n - 1}$$

Standard Deviation (Sample):

$$s = \sqrt{\frac{\sum f(x - \bar{x})^2}{n - 1}}$$

Measures of Center:**Table with Class Intervals:**

<i>Year</i>	<i>Percent Change, x</i>	<i>Frequency (f)</i>	<i>f * x</i>
1983	2.1%	1	2.1%
1976	3.0%	1	3.0%
1984	3.8%	1	3.8%
1982	4.1%	1	4.1%
1977	6.3%	1	6.3%
1981	7.8%	1	7.8%
1975	8.5%	1	8.5%
1980	8.6%	1	8.6%
1978	9.9%	1	9.9%
1979	11.0%	1	11.0%
1974	14.3%	1	14.3%
		$\Sigma f = 11$	$\Sigma fx = 79.4\%$

Mean:

$$\bar{x} = \frac{\Sigma Fx}{\Sigma F} = \frac{79.4}{11} = 7.181818\% \cong 7.22\%$$

The average percent change of the CPI for All Foods over the years 1974-1984 is 7.22%.

Median:

Being an odd number of data points, the median will be the 6th data point.

~~2.1%, 3.0%, 3.8%, 4.1%, 6.3%, 7.8%, 8.5%, 8.6%, 9.9%, 11.0%, 14.3%~~

Median = 7.8%

In ascending order, the median percent change of the CPI for All Foods over the years 1974-1984 is 7.8%

Mode:

Because all data points are different and none repeat, there is no mode for this dataset.

Midrange:

$$\text{Midrange} = \frac{\text{min} + \text{max}}{2}$$

The minimum value is the year 1983 with 2.1% and the maximum value is the year 1974 with 14.3%.

$$\text{Midrange} = \frac{2.1\% + 14.3\%}{2} = \frac{16.4\%}{2} = 8.2\%$$

The average value of the minimum and the maximum changes of the CPI for All Foods for the years 1974-1984 is 8.2%

Measures of Spread:**Range:**

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

The minimum value is the year 1983 with 2.1% and the maximum value is the year 1974 with 14.3%.

$$\text{Range} = 14.3\% - 2.1\% = 12.2\%$$

The difference between the maximum change and the minimum change of the CPI for All Foods over the years 1974-1984 is 12.2%

Variance (Sample)

$$\Sigma fx = 79.4\%$$

$$\bar{x} = 7.2\%$$

$$n = 11$$

$$s^2 = \frac{\Sigma (x - \bar{x})^2}{n - 1}$$

Percent Change	Frequency	$x - \bar{x}$	$(x - \bar{x})^2$	$f(x - \bar{x})^2$
2.1%	1	-5.118181	26.01	26.195785
3.0%	1	-4.218181	17.64	17.793057
3.8%	1	-3.418181	11.56	11.683966
4.1%	1	-3.118181	9.61	9.723057
6.3%	1	-0.918181	0.81	0.843057
7.8%	1	0.581818	0.36	0.338512
8.5%	1	1.281818	1.69	1.643057

8.6%	1	1.381818	1.96	1.909421
9.9%	1	2.681818	7.29	7.192148
11.0%	1	3.781818	14.44	14.302148
14.3%	1	7.081818	50.41	50.152148
	$\Sigma f = 11$			$\Sigma f(x - \bar{x})^2$ $= 141.776363$

$$s^2 = \frac{141.776363}{11 - 1} = \frac{141.776363}{10} = 14.177636\%^2 \cong 14.18\%^2$$

The sample variance of the CPI for All Foods over the years 1974-1984 is approximately 14.18%

Standard Deviation (Sample):

$$\Sigma f(x - \bar{x})^2 = 141.776363$$

$$n = 11$$

$$s = \sqrt{\frac{\Sigma f(x - \bar{x})^2}{n - 1}}$$

$$= \sqrt{\frac{141.776363}{11 - 1}} = \sqrt{\frac{141.776363}{10}}$$

$$= \sqrt{14.177636} = 3.765320 \cong 3.77\%$$

The sample standard deviation of the CPI for All Foods over the years 1974-1984 is 3.77%

The variation of the dataset values, as a measure of the standard deviation, is about 3.77%.

Sorted in Ascending Order: 2.1%, 3.0%, 3.8%, 4.1%, 6.3%, 7.8%, 8.5%, 8.6%, 9.9%, 11.0%, 14.3%

$$n = 11$$

Lower Quartile:

$$\text{Lower Quartile} = \frac{1}{4}n = \frac{1}{4} * 11 = 2.75$$

Because the integer is a decimal, we round up from 2.75 to 3. The 3rd data value in ascending order is 3.8%. Meaning...

$$\text{Lower Quartile} = 3.8\%$$

Third Quartile:

$$\text{Upper Quartile} = \frac{3}{4}n = \frac{3}{4} * 11 = 8.25$$

Because the integer is a decimal, we round up from 8.25 to 9. The 9th data value in ascending order is 9.9%. Meaning...

$$\text{Upper Quartile} = 9.9\%$$

To find the Interquartile Range (IQR) you must find the difference between the Upper and Lower Quartiles.

$$IQR = \text{Upper Quartile} - \text{Lower Quartile}$$

$$IQR = 9.9\% - 3.8\% = 6.1\%$$

The IQR of the CPI for All Foods over the years 1974-1984 is 6.1%

6.1% account for the variation in 50% of the percent changes of the CPI for All Foods over the years 1974-1984.

Measures of Position:

The Five Number summary of the dataset is:

Minimum: 2.1%

The minimum percent change of the CPI for All Foods over the years 1974-1984 is below 2.1%.

Lower Quartile: 3.8%

25% of the percent changes of the CPI for All Foods over the years 1974-1984 is below 3.8%

Middle Quartile (Median): 7.8%

Second Quartile:

$$\text{Second Quartile (Median)} = \frac{2}{4}n = \frac{2}{4} * 11 = 5.5$$

Because the integer is a decimal, we round up from 5.5 to 6. The 6th data value in ascending order is 7.8%. Meaning...

Second Quartile (Median) = 7.8%

50% of the percent changes of the CPI for All Foods over the years 1974-1984 is below 7.8%

Upper Quartile: 9.9%

75% of the percent changes of the CPI for All Foods over the years 1974-1984 is below 9.9%

Maximum: 14.3%

The maximum percent change of the CPI for All Foods over the years 1974-1984 is 14.3%

References (MLA Style)

U.S. Bureau of Labor Statistics. "Consumer Price Index." *Bls.gov*, U.S. Bureau of Labor Statistics, 2025, www.bls.gov/cpi/.

Chukwuemeka, Samuel. "Descriptive Statistics." *Descriptive Statistics Projects*, statistical-reasoning.appspot.com/. Accessed 30 Apr. 2025.

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